Model Validation & Hyperparameter Tuning Report

California Housing Price Prediction — AI/ML Task 3

1. Overview

This report summarizes the model validation and tuning workflow carried out on the California Housing dataset (20,640 rows, 8 numeric features, target variable: median house price). Two regression models were trained and compared — Linear Regression and Decision Tree Regressor — with the Decision Tree subjected to overfitting analysis, k-fold cross-validation, and hyperparameter tuning via GridSearchCV.

2. Overfitting Analysis

2.1 Train vs. Test Score Gap

The default Decision Tree Regressor (trained with no depth or leaf-size restrictions) achieved a training R^2 score of **1.000** but a test R^2 score of only **0.622**. A perfect training score combined with a much lower test score is a textbook indicator of overfitting: the tree grew deep enough to memorize every training sample (including noise), rather than learning the underlying relationship between features and price.

2.2 Cross-Validation Confirms Instability

5-fold cross-validation on the full dataset (scoring = R^2) produced the following per-fold scores: 0.271, 0.414, 0.439, 0.236, and 0.419, with an **average CV R^2 of 0.356**. Two observations follow from this:

- **Large variance across folds** (0.236 to 0.439) shows the unconstrained tree's performance depends heavily on which subset of data it is trained on — a hallmark of a high-variance, overfit model.
- **The average CV score (0.356) is far below the single test-split score (0.622)**, meaning the original 80/20 split happened to be a relatively favorable one for this model, and that single-split score overstates how well the model generalizes.

2.3 Linear Regression as a Baseline

For comparison, Linear Regression (a low-variance, high-bias model) achieved RMSE = 0.746 and R^2 = 0.576 on the test set, with no train/test gap to speak of. This confirms the dataset has a learnable signal (R^2 well above zero) but also that a simple linear model underfits, leaving room for a more flexible model — provided that flexibility is controlled.

3. Tuning Approach

To address the overfitting identified above, the Decision Tree's complexity-controlling hyperparameters were tuned using **GridSearchCV** with 5-fold cross-validation and R^2 as the scoring metric. The search grid covered:

- **max_depth**: [5, 10, 15, 20] — caps how many splits deep the tree can grow, directly limiting its ability to fit noise.
- **min_samples_split**: [2, 5, 10] — the minimum number of samples a node must have before it can be split further.
- **min_samples_leaf**: [1, 2, 4] — the minimum number of samples allowed in a leaf node, preventing the tree from creating leaves for single outlier points.

This produced 4 × 3 × 3 = 36 hyperparameter combinations, each evaluated with 5-fold CV (180 model fits total). The combination that maximized average cross-validated R^2 on the training set was:

max_depth = 10, min_samples_leaf = 4, min_samples_split = 2

Intuitively, this configuration limits the tree to a moderate depth of 10 (down from an unbounded depth that previously reached 100% training accuracy) and requires each leaf to represent at least 4 training samples, which smooths out predictions and reduces sensitivity to individual data points.

4. Model Comparison & Final Model Justification

Model	RMSE	R^2 Score
Linear Regression	0.7456	0.5758
Decision Tree (Before Tuning)	0.7037	0.6221
Decision Tree (After Tuning)	0.6391	0.6883

4.1 Impact of Tuning

Tuning reduced RMSE from 0.7037 to 0.6391 (a ~9.2% reduction in prediction error) and increased R^2 from 0.6221 to 0.6883 (an improvement of roughly 6.6 percentage points in explained variance). The tuned tree also no longer exhibits the 1.000 vs 0.622 train/test gap seen earlier, since its depth and leaf-size constraints prevent it from perfectly memorizing the training data.

4.2 Final Model Recommendation

The **tuned Decision Tree Regressor (max_depth=10, min_samples_leaf=4, min_samples_split=2)** is recommended as the final model for this task, for the following reasons:

- It achieves the **lowest RMSE and highest R^2** of all models tested on the held-out test set.
- It directly resolves the overfitting identified in Section 2 by constraining tree depth and leaf size, rather than allowing unlimited growth.
- It outperforms Linear Regression, indicating the dataset benefits from the non-linear splits a tree can capture, as long as that flexibility is regularized.

4.3 Limitations & Next Steps

An R^2 of ~0.69 still leaves roughly 31% of the variance in housing prices unexplained. Suggested next steps include re-running cross-validation on the *tuned* model (rather than just the train/test split) to confirm the improvement holds across folds, engineering additional features (e.g. rooms per household, population density), and evaluating ensemble methods such as Random Forest or Gradient Boosting, which typically reduce variance further while retaining the Decision Tree's ability to model non-linear relationships.